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FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

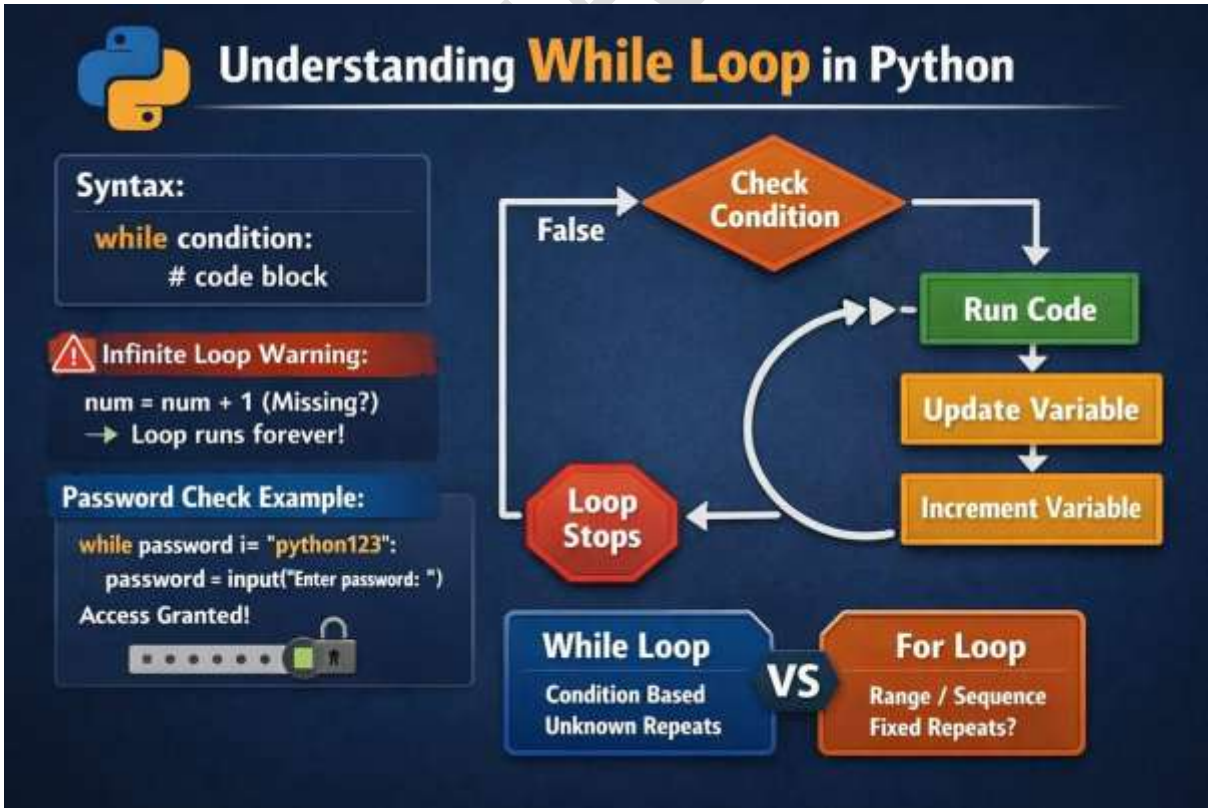
Classes : Offline & Online

Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Lesson 23: While Loop in Python



Learning Objectives

By the end of this lesson, students will be able to:

- ✅ Understand what a **While Loop** is.
- ✅ Learn when to use a while loop.
- ✅ Write programs using while loops.
- ✅ Understand loop conditions and counters.
- ✅ Avoid infinite loops.
- ✅ Solve simple real-life problems using repetition.

What is a While Loop?

A **While Loop** is used to repeat a block of code **as long as a condition is True**.

Unlike a **for loop**, a **while loop** is used when we **do not know in advance how many times** the loop should repeat.

Real-Life Example

Imagine filling a water bottle.

- Keep pouring water **while** the bottle is **not full**.
- Stop when the bottle becomes full.

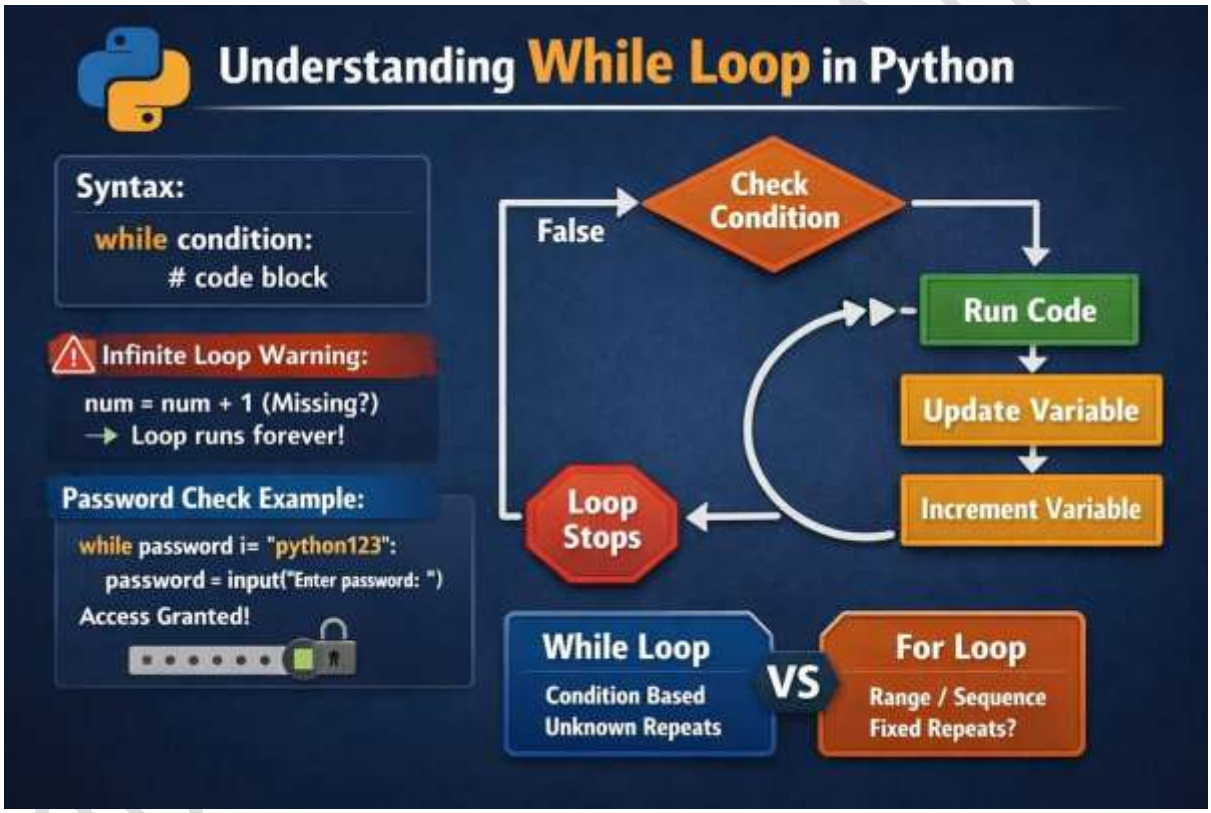
Similarly, Python repeats the code **while the condition remains True**.

Real-Life Example of a While Loop

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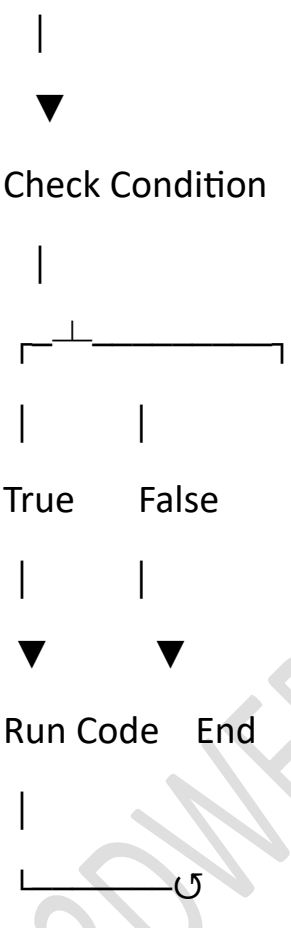
Why Do We Use While Loops?

While loops help us:

- 🔁 Repeat tasks until a condition changes.
- 🎮 Create games that continue until the player quits.

- 🔑 Ask for a password until the correct one is entered.
- 📊 Build menus that repeat until the user exits.
- 🤖 Automate repetitive tasks.

How a While Loop Works



Syntax of While Loop

```
while condition:
    statement
```

The loop continues **until the condition becomes False**.

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Example 1

Print numbers from **1 to 5**.

```
i = 1


while i <= 5:
    print(i)
    i += 1
```

Output

```
1
2
3
4
5
```

Understanding the Program

```
i = 1
```

Starts counting from **1**

```
while i <= 5:
```

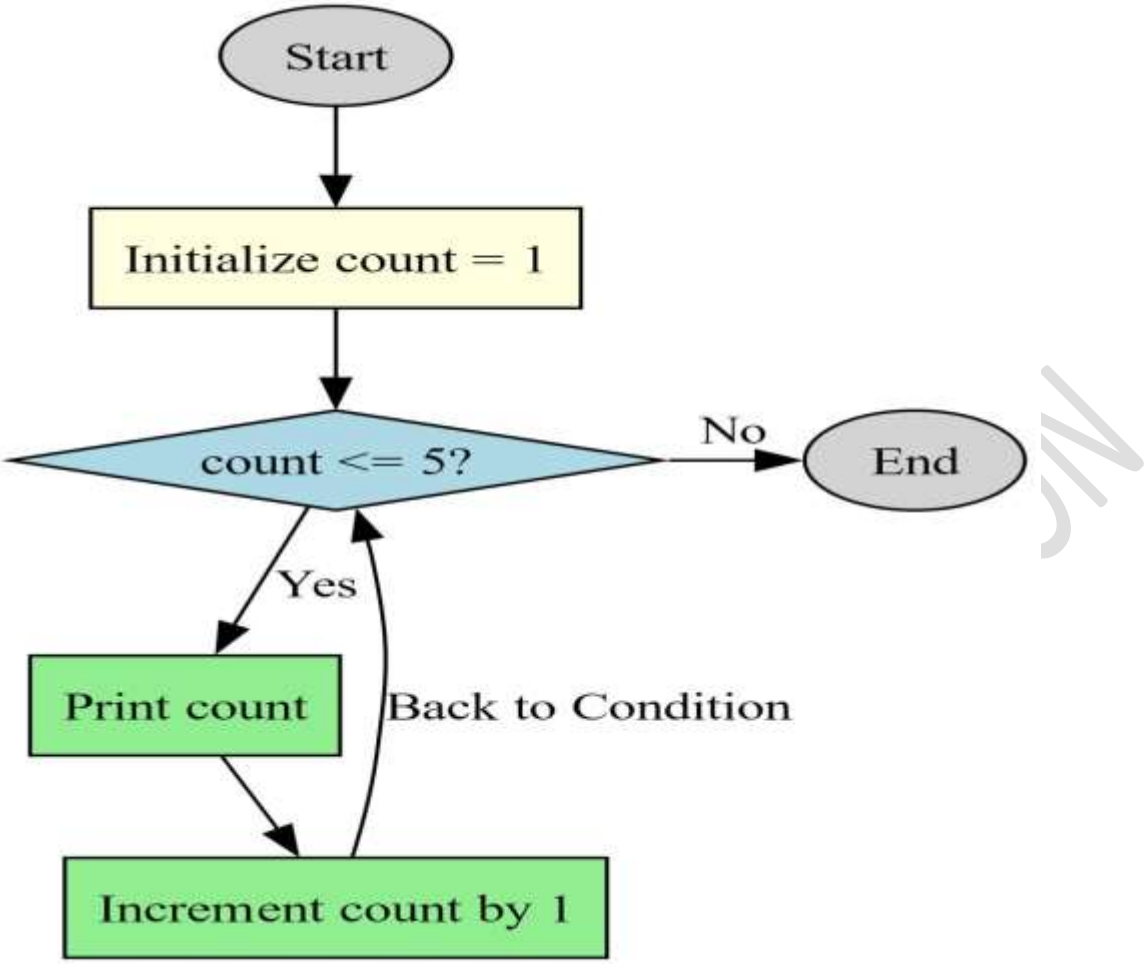
Repeat while **i is less than or equal to 5**

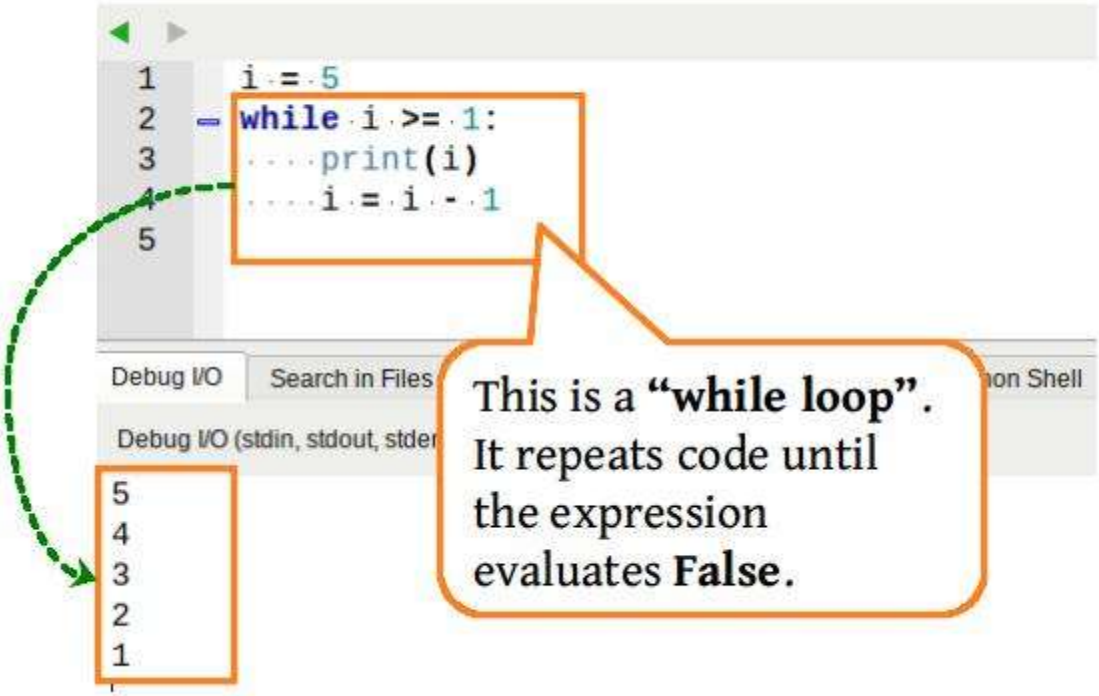
```
i += 1
```

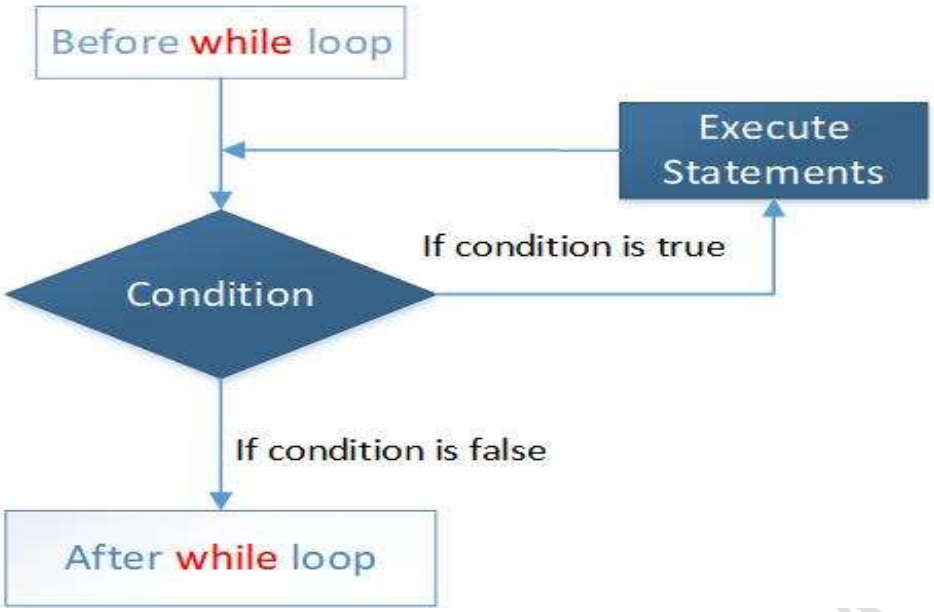
Increase i by **1** after every repetition.

While Loop Flowchart

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Start
Initialize count = 1
count <= 5?
No
End
Yes
Print count
Back to Condition
Increment count by 1


1 i = 5
2 while i >= 1:
3 print(i)
4 i = i - 1
5
Debug I/O
Search in Files
Debug I/O (stdin, stdout, stder
5
4
3
2
1
This is a "while loop". It repeats code until the expression evaluates False.



Example 2

Print "Hello Python" five times.

```
count = 1


while count <= 5:
    print("Hello Python")
    count += 1
```

Output

```
Hello Python
Hello Python
Hello Python
Hello Python
Hello Python
```

Example 3

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Print Even Numbers

number = 2

```
while number <= 10:
   print(number)
   number += 2
```

Output

2

4

6

8

10

Example 4

Print Odd Numbers

number = 1

```
while number <= 9:
   print(number)
   number += 2
```

Output

1

3

5

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7

9

Example 5

Multiplication Table of 5

```
i = 1

while i <= 10:
    print("5 x", i, "=", 5 * i)
    i += 1
```

Example 6

Sum of Numbers from 1 to 5

```
total = 0
i = 1

while i <= 5:
    total += i
    i += 1

print("Total =", total)
```

Output

Total = 15

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Example 7

Countdown

```
count = 5


while count >= 1:
    print(count)
    count -= 1


print("Blast Off!")
```

Output

5

4

3

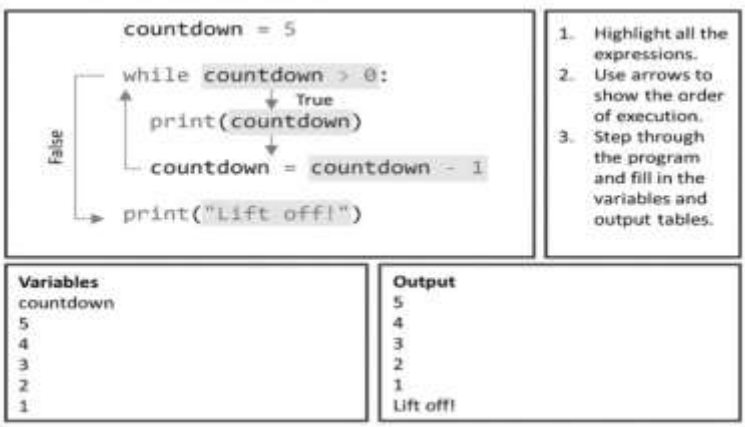
2

1

Blast Off!

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Countdown Using While Loop



Infinite Loop

An **Infinite Loop** keeps running forever because the condition never becomes False.

Wrong Example

```
i = 1

while i <= 5:
    print(i)
```

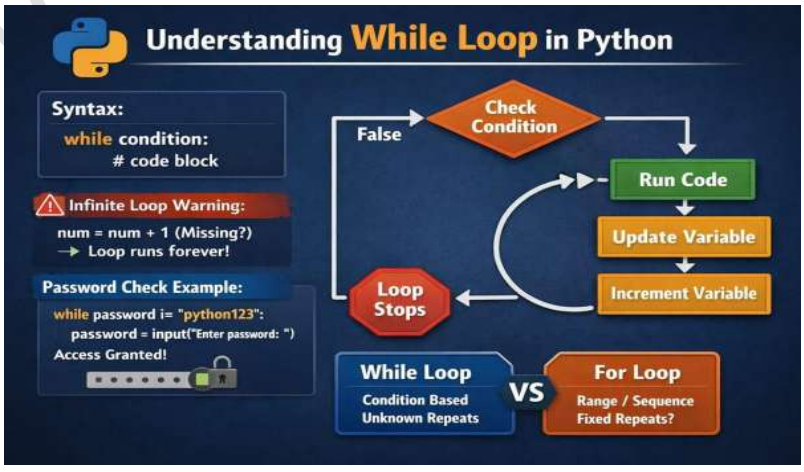
❌ i never changes, so the loop never ends.

Correct Example

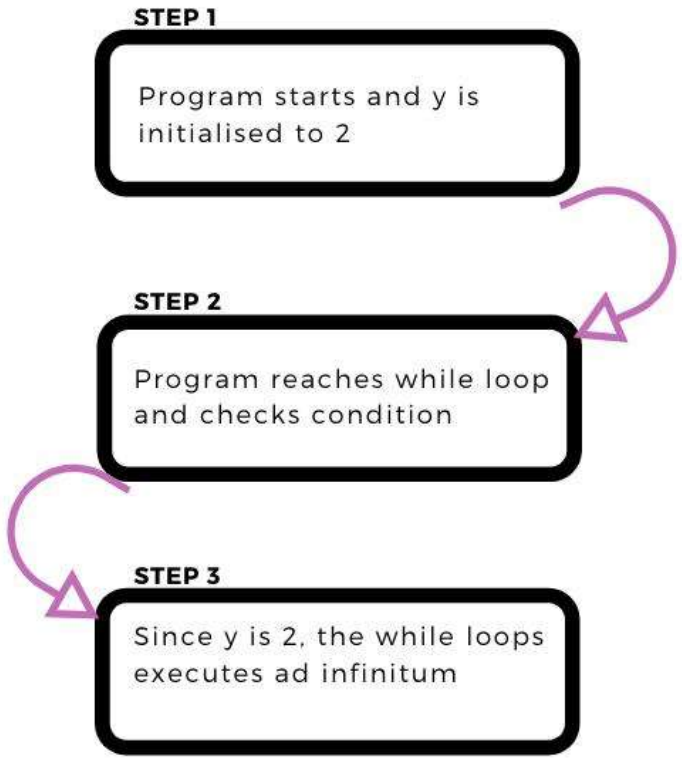
```
i = 1

while i <= 5:
    print(i)
    i += 1
```

Infinite Loop Warning



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Practical Session

Practical Activity 1

Print Numbers 1 to 10

```
i = 1


while i <= 10:
    print(i)
    i += 1
```

Practical Activity 2

Print Your Name 5 Times

```
name = input("Enter Your Name: ")

count = 1

while count <= 5:
    print(name)
    count += 1
```

Practical Activity 3

Multiplication Table

```
number = int(input("Enter Number: "))

i = 1

while i <= 10:
    print(number, "x", i, "=", number * i)
    i += 1
```

Practical Activity 4

Print Even Numbers

```
number = 2

while number <= 20:
    print(number)
```

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```
    number += 2
```

Practical Activity 5

Print Odd Numbers

```
number = 1

while number <= 19:
    print(number)
    number += 2
```

Practical Activity 6

Sum of Numbers

```
total = 0
i = 1

while i <= 10:
    total += i
    i += 1

print(total)
```

Practical Activity 7

Countdown Timer

```
count = 10
```

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```
while count >= 1:
    print(count)
    count -= 1

print("Time Up!")
```

Practical Activity 8

Password Checker

```
password = ""

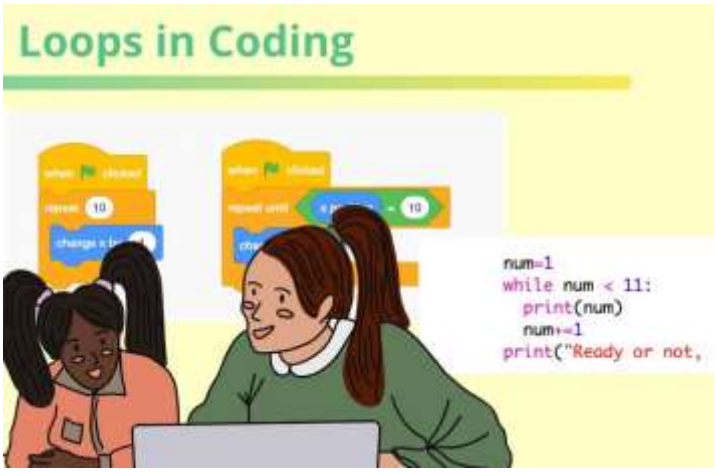
while password != "python123":
    password = input("Enter Password: ")

print("Access Granted!")
```

Classroom Coding Activity



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Activity 1

Print numbers from **1 to 20**.

Activity 2

Print even numbers from **2 to 50**.

Activity 3

Display the multiplication table of **7**.

Activity 4

Create a countdown from **20** to **1**.

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Difference Between For Loop and While Loop

For Loop	While Loop
Used when the number of repetitions is known.	Used when the number of repetitions is unknown.
Uses range() often.	Uses a condition.

For Loop	While Loop
Simpler for counting.	Better for repeated checking.

Real-Life Applications

Situation	While Loop Use
ATM PIN Entry	Repeat until the correct PIN is entered
Game	Continue until the player quits
Login Screen	Ask for the password until it is correct
Mobile Lock	Retry until unlocked
Online Quiz	Continue until all questions are answered
Menu Program	Show options until Exit is selected

Common Mistakes

❌ **Forgetting to Update the Counter**

Wrong

```
i = 1

while i <= 5:
    print(i)
```

This creates an **Infinite Loop**.

✅ **Correct**

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```
i = 1

while i <= 5:
    print(i)
    i += 1
```

❌ **Missing Colon (:)**

Wrong

```
while i <= 5
```

✅ **Correct**

```
while i <= 5:
```

❌ **Incorrect Indentation**

Wrong

```
while i <= 5:
print(i)
```

✅ **Correct**

```
while i <= 5:
    print(i)
```

Best Practices

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- ✓ Always update the loop variable.
- ✓ Write simple conditions.
- ✓ Use meaningful variable names.
- ✓ Test your program with different inputs.
- ✓ Add comments to explain your code.

Keywords

- **Loop** – Repeats a block of code.
- **While Loop** – Repeats while a condition is True.
- **Condition** – A True or False expression.
- **Counter** – A variable that keeps track of repetitions.
- **Infinite Loop** – A loop that never ends.

Lesson Summary

- A **While Loop** repeats code while a condition is True.
- It is useful when we do not know how many times the loop should run.
- The loop stops when the condition becomes False.
- The counter variable must be updated to avoid infinite loops.
- While loops are commonly used in login systems, games, and menus.

Quiz

Multiple Choice Questions

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1. Which keyword starts a while loop?

A) for

B) if

C) while

D) range

✅ **Answer: C**

2. A while loop repeats as long as:

A) The condition is True

B) The condition is False

C) The program ends

D) The computer is on

✅ **Answer: A**

3. What happens if you forget to increase the counter?

A) The program runs faster.

B) The program stops immediately.

C) An infinite loop may occur.

D) Nothing happens.

✅ **Answer: C**

4. Which loop is better when the number of repetitions is unknown?

A) for

B) while

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C) if

D) elif

✅ **Answer: B**

5. What will this code print?

```
i = 1

while i <= 3:
    print(i)
    i += 1
```

A) 1 2 3

B) 0 1 2

C) 1 2

D) Infinite Loop

✅ **Answer: A**

Homework

1. Write a program to print numbers from **1 to 30** using a while loop.
2. Write a program to print even numbers from **2 to 20**.
3. Write a program to display the multiplication table of **9**.
4. Write a program to calculate the sum of numbers from **1 to 20**.
5. Write a countdown program from **15** to **1**.
6. Create a password checker using a while loop.
7. Write a program to print the cubes of numbers from **1 to 10**.

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